Pakistan Engineering Council

Constituted under Pakistan Engineering Council Act, 1976 enacted by the Parliament)

Ataturk Avenue (East),
G-5/2, Islamabad

Engineering Accreditation Department

PEC/EAD/EQEEP/2025/04
July 14, 2025

The Vice Chancellors/ Rectors/ Heads
of all HEIs Offering Engineering Programs in Pakistan

Subject: **Engagement of HEIs for Input on Integration of International Certifications into Engineering Degree Programs**

Respected Sir/ Madam,

The Pakistan Engineering Council (PEC), under the leadership of **Chairman Engr. Waseem Nazir**, is implementing strategic reforms to modernize engineering education in Pakistan. These efforts focus on curriculum enhancement, practical skill development and alignment with both national priorities and global industry standards. Key initiatives include the "Pathway to Practice" model to support graduate employability, the "Brain Gain Portal" to engage the global Pakistani engineering diaspora, and Generative AI Training Program targeting 15,000 engineers annually. These programs reflect PEC's commitment to innovation, capacity building and strengthening the global competitiveness of Pakistani engineers.

As part of these ongoing efforts, PEC has constituted a high-level **Task Force (TF)** under the initiative titled *"**Enriching Quality Engineering Education in Pakistan**"*, which has mandate to develop mechanisms for integrating internationally recognized skill-based certifications into undergraduate engineering curricula. These certifications are intended to enhance students' practical competencies and employability, offering flexible learning pathways aligned with global industry benchmarks.

During its inaugural meeting held on 21st May 2025, the Task Force reviewed a range of internationally recognized certifications and proposed that students might be allowed to undertake such certifications, subject to approval by PEC and the respective institutions, in lieu of 6–12 credit hours of technical or non-technical elective coursework, through a defined mechanism. Furthermore, the Task Force assigned responsibilities to its members to engage with certification bodies and explore the possibility of negotiating discounted rates for Pakistani students.

To ensure that the proposed framework is relevant, practical, and discipline-specific, TF seeks your esteemed institution's valuable input. A **sample/ proposed list of certifications** is enclosed as ***Annex-A*** for your reference. You are requested to review this list and:

- Recommend additional certifications appropriate for your engineering programs/ disciplines and industry engagement.

Direct: 051-9204466, Fax: 051-9219050, 9214488
Website: http://www.pec.org.pk Email: junaid.khan@pec.org.pk

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- Suggest any necessary adjustments in credit hour allocations to support integration without disrupting the academic structure.
- Please provide details of any existing MoUs/ Agreements between your institution and any certification-awarding institute or organization (if any).

This initiative is proposed for implementation from the **Fall Semester 2026**, subject to approval by PEC's **Engineering Accreditation Board (EAB)**, **Governing Body**, and other relevant forums. Your institution's engagement in this consultative process is critical to the success of this national initiative.

We kindly request your feedback within **two weeks** from the date of this communication to enable timely consideration and finalization. Your collaboration in advancing engineering education in Pakistan is highly valued.

With Warm Regards,

Engr. Aijaz Hussain Shah
Additional Registrar/
HoD EAD

Encl: Annex-A: Sample/ Proposed List of International Certifications

Copy to:

i. The Convener, PEC Engineering Accreditation Board (EAB)
ii. Lead and Members of the TF on Enriching Quality Engineering Education in Pakistan
iii. PS to Chairman/ Registrar, PEC

Direct: 051-9204466, Fax: 051-9219050, 9214488
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Annex- A

Sample/ Proposed list of certifications

Sr#	Certification Name	International Rating/ 5	Proposed Cr. Hr. Equivalency	Course Duration	Website Link
Engineering Discipline: Electrical/Telecom Engineering					
1)	Cisco Certified Network Associate (CCNA)	4.8			
2)	Microsoft Power BI Data Analyst	4.6			
3)	Adobe Certified Expert (ACE)	4.7			
4)	Google IT Support Professional	4.6			
5)	Certified Blockchain Developer	4.7			
6)	VMware Certified Professional - Data Center Virtualization (VCP-DCV)	4.8			
7)	Certified SOLIDWORKS Professional (CSWP)	4.7			
Engineering Discipline: Information Security					
1)	CISSP	4.8			
2)	CISM	4.8			
3)	CSA by SOC	4.6			
4)	Blue Team Level 1	4.8			
5)	Red Hat Certified Engineer	4.8			
6)	Microsoft Certified: Azure Fundamentals	4.7			
7)	Cisco Certified Network Associate (CCNA)	4.7			
8)	Microsoft Power BI Data Analyst	4.5			
Engineering Discipline: Computer Science/ Software Engineering					
1)	Google Professional Machine Learning Engineer	4.8			
2)	AWS Certified Machine Learning – Specialty	4.7			
3)	Certified AI & Machine Learning Professional	4.5			
4)	NVIDIA Deep Learning AI Certification	4.8			
5)	AWS Certified DevOps Engineer	4.8			
6)	Oracle Certified Java Programmer	4.5			
7)	AWS Certified DevOps Engineer	4.8			